Context length, conversation length, and cross-session memory as sources of factual unreliability in LLM assistants

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Summary

The question. LLM assistants sometimes get facts wrong, or contradict themselves, as inputs and conversations grow longer or when they draw on earlier sessions. This review asks how far such failures relate to three things:

- context length within a single input;
- conversation length within a session;
- cross-session memory, meaning the systems that store, retrieve and update information between sessions.

It also asks whether a given failure can be attributed to one mechanism rather than another. The approved boundaries treat three premises as claims to test, not assumptions: that length degrades reliability, that the two mechanisms are empirically separable, and that effective context is smaller than the advertised window.

What the controlled evidence shows.

1. Within a single input, longer context does degrade reliability, even when the needed evidence is held fixed.

- Two padding designs show this directly. Levy et al. (2024) padded inputs with duplicated, task-similar or unrelated text. Du et al. (2025) padded with whitespace, or with attention-masked tokens after confirming that the model could recite the evidence. Both found accuracy falling as length grew.
- The effect is moderate to large in open models of 70B parameters or smaller, at 30K tokens or fewer. It is smaller, but not absent, in closed frontier models.
- Distractors that resemble the target, and low lexical overlap between question and evidence, make the decline much steeper (NoLiMa; Context Rot; Jin et al. 2025).
- Position effects ("lost in the middle") are real but conditional. They are clearest at short absolute lengths and when the input fills less than about half the window. Beyond that, they shift toward a simple recency gradient (Veseli et al. 2025).

2. Effective context is smaller than the advertised window on almost every non-trivial test, but the size of the gap depends on how "effective" is defined.

- RULER defines it with an absolute threshold. NoLiMa defines it relative to each model's own short-context score. BABILong expresses it as a share of the window.
- These definitions give numbers that are not interchangeable: from "more than 128K" to "2K or less" for the same generation of models.
- Single-needle retrieval tests saturate near 100% and say little about factual reliability on realistic tasks (RULER; HELMET; the Gemini 1.5 report itself).

3. Within a session, multi-turn interaction causes large drops, but mostly not because of turn count or token count as such.

- When a request is revealed piece by piece over several turns, accuracy falls by about 35–39% on average. Giving the same pieces in one message keeps about 95% of single-turn performance (Laban et al. 2025).
- Designs that control what the model sees point to four causes:
  - **underspecification:** the model answers before it has the whole request;
  - **conditioning on its own earlier mistakes:** self-generated errors carried forward in the history (MT-Eval; Sinha et al.);
  - **interference from superseded facts:** this persists even at fixed total length (PI-LLM);
  - **social pressure in user turns:** for example, giving way when challenged (FlipFlop).
- Pure length still contributes a smaller residual decline. Neutral filler turns can be harmless.

4. Cross-session memory fails in ways that are distinct from long-context failures. The best-supported memory failure modes are:

- updating and resolving contradictions;
- temporal reasoning across sessions;

- aggregating facts spread over several sessions;
- answering false-premise questions instead of abstaining;
- write-stage loss (facts never stored, or stored with the wrong date);
- applying a stored fact after it has stopped being true.

When memory pipelines are split into stages, retrieval failures dominate in some studies (Yuan et al. 2026). In others, failures split roughly evenly between retrieval and reading (LongMemEval).

5. The two mechanisms can be separated in controlled experiments, but they are nested rather than parallel. A retrieved memory is itself placed into a context that the model must read. Oracle-retrieval conditions, length-matched controls, stage-wise decompositions and route substitution (retrieval versus long context, per query) each isolate one factor. None isolates all of them.

What this means for attribution.

- Attribution holds at the population level: under a named controlled contrast, on a stated benchmark.
- It rarely holds for a single failure in a deployed assistant. The reasons:
  - the pipelines are undocumented;
  - outputs vary from run to run;
  - judge and annotation noise is often as large as the effects being attributed;
  - the model's own explanations are not valid evidence of which mechanism failed.

Product reports. Vendor documentation and user reports are anecdotal evidence, used here only as illustrations.

- Vendors' own documentation concedes the main limits:
  - memory is selective;
  - deleting a memory does not take effect immediately;
  - provenance displays are incomplete;
  - long chats are summarised as they near the window limit;
  - effective context is smaller than the nominal window.
- User reports describe failures that resemble the controlled failure modes. They cannot establish which mechanism caused any particular failure.

> **Verdict on the three premises.** (1) Longer context and longer conversations degrade factual reliability: *supported, with boundary conditions*. The effect is task- and model-dependent, and in multi-turn settings it is driven more by interaction structure than by length. (2) The two mechanisms are empirically separable: *supported in controlled settings, not for individual deployed failures*. (3) Effective utilisation is smaller than the advertised window: *supported on every non-trivial benchmark*, though its magnitude depends on the definition used.

1. Scope and method

1.1 What was reviewed

This is a structured narrative evidence review. It is not systematic, so it makes no claim to have found every relevant study.

In scope: text-based LLMs and LLM assistant systems, including systems with retrieval-based or other cross-session memory modules; work from 2020 to the date of search (3 November 2026); and three topics: processing of long inputs within a session, multi-turn interaction within a session (what "conversation length" means here), and cross-session memory (how information is stored, retrieved, updated and reasoned about in time).

Mitigation techniques (retrieval, summarisation, compaction) appear only where they help explain or separate the mechanisms. They are not assessed or recommended.

Out of scope: multimodal-only work; factual errors that come from training-data knowledge and are unrelated to context or memory; and any comparison or ranking of commercial products.

1.2 Search and appraisal

Evidence was gathered in five parallel sweeps: single long inputs; multi-turn interaction within one session; cross-session memory; evaluation designs that separate the mechanisms and the limits of attribution; and reports about commercial products. Each sweep searched arXiv, the ACL Anthology, OpenReview and conference proceedings, and deliberately looked for negative results, boundary conditions and competing explanations. For each study that drives a conclusion, the review recorded design, models and versions, context lengths, comparators and controls, metrics, uncertainty reporting, confounds, and level of access (condensed in Section 10). Numbers were checked against primary sources where accessible; values available only from figures or secondary reports are marked as such.

1.3 Labels used in this report

Publication status. Each study is labelled peer-reviewed (with venue), preprint, technical report or vendor blog. Several 2025–2026 papers were available only as preprints. Their findings are used but flagged, and they do not carry a conclusion on their own.

Strength of conclusion.

Label	Meaning
Strong	Several independent studies with controlled designs agree.
Moderate	Consistent controlled evidence, but from few groups, few models or synthetic tasks only.
Limited	A single study, preprint-only support, abstract-only access, or a substantial design confound.

Evidence classes.

- *Controlled evaluations* are the primary evidence.
- *Product reports* (vendor documentation, vendor announcements, journalism, forum posts) form a separate anecdotal class. Section 8 covers them. They are never treated as evidence that a specific mechanism caused a failure.

1.4 The two mechanisms, and where the line between them blurs

Mechanism (a): within-session context processing. Everything the model needs is in its context window. The failure is in how the model uses that context.

Mechanism (b): cross-session memory. Information from earlier sessions has to be written to a store, indexed, retrieved, inserted into the current context, and kept up to date. The failure can happen at any of these steps.

These are not mutually exclusive.

- A retrieved memory ends up inside a context window. So a memory failure can be a reading failure, mechanism (a) operating inside (b).
- Systems that summarise or "compact" a long conversation within one session use the same lossy operations as memory systems. They do so without crossing a session boundary.

Following the approved boundaries, such cases are assigned according to how the study or system describes its own mechanism. Where that assignment is ambiguous, the report says so.

2. Measuring "effective context"

No definition of effective context utilisation has been agreed. The review therefore reports each source's definition instead of imposing one. This matters because the definitions produce numbers that cannot be compared with each other.

Source	Operational definition	What it implies
RULER (Hsieh et al., COLM 2024)	Longest tested length at which the model's average over 13 synthetic tasks stays above a fixed *absolute* threshold: Llama-2-7B's score at 4K, 85.6%.	Strong models can reach "more than 128K". The threshold is arbitrary and tied to one reference model.
NoLiMa (Modarressi et al., ICML 2025)	Longest length at which the score stays above 85% of the model's *own* short-context score, on needles with minimal lexical overlap with the question.	Most of 13 models with windows of 128K or more fall to 2K or less. The relative criterion is strict.
BABILong (Kuratov et al., NeurIPS 2024 D&B)	Share of the window over which bAbI reasoning remains intact when facts are hidden in book text.	Popular models use about 10–20% of their context. The figure depends on the task.
An et al. (ICLR 2025)	Effective length as a share of *training* length, attributed to how rarely long relative positions occur during training.	Open models typically use no more than about half their training length.
HELMET (Yen et al., ICLR 2025)	Declines to give one number. Measures performance by task category at 8K–128K.	Argues that effective context is task-specific.

Practical consequence. A statement such as "model X has an effective context of N tokens" means something only when the definition and task set are stated with it. A finding that one definition places effective length well below the window is not contradicted by another definition placing it near the window. They measure different things.

3. Q1a: Reliability within a single long context

Intuition. A model reading a long input has to do two things. First it must find the information that matters. Then it must use that information correctly, with everything else still in view. A failure could come from either step. It could also come from the input simply being long, even when nothing in it is misleading. The studies below try to pull these three possibilities apart.

3.1 Length itself, with the evidence held fixed

Design. The cleanest designs keep the task and its evidence identical and change only how much extra material surrounds it.

Levy, Jacoby and Goldberg (ACL 2024): FLenQA

- **Setup:**
  - Three true/false reasoning tasks, 100 base instances each.
  - Two key paragraphs were padded to about 250–3,000 tokens with one of three kinds of filler:
    - duplicate copies of the key paragraphs, which adds almost no new information;
    - paragraphs from other instances of the same task;
    - unrelated BookCorpus text.
  - Key paragraphs were placed at the start, middle or end, or dispersed.
- **Models:** GPT-4, GPT-3.5, Gemini Pro, Mistral Medium and Mixtral 8x7B, with and without chain-of-thought (CoT).
- **Result:**
  - Mean accuracy fell from 0.92 at about 250 tokens to 0.68 at about 3,000 tokens. That is far below any of these models' limits.
  - The fall appeared even with duplicate padding.
  - Unrelated text usually hurt more than similar text.
  - Dispersed evidence was worst. Evidence at the end was generally best.
  - CoT added a roughly constant gain but did not close the gap (GPT-4 was the exception).
  - As inputs grew, models refused more often, drifted toward answering "False", and their reasoning covered fewer of the relevant facts.
- **Limitations:** Lengths are short by current standards. Binary labels mean the drift toward "False" interacts with the accuracy measure. The closed models were tested at undisclosed dates.

Du et al. (Findings of EMNLP 2025): "Context Length Alone Hurts…"

- **Setup:** This design goes further. Each prompt has evidence, then distraction, then the question.
  - Retrieval is measured separately by asking the model to recite the evidence and question exactly. Accuracy can then be analysed only on items where retrieval succeeded.
  - Three kinds of distraction:
    - Paul Graham essays;
    - whitespace;
    - attention-masked tokens, which advance positional indices but carry no visible content.
- **Models:** Llama-3.1-8B, Mistral-7B, GPT-4o, a Claude 3.x model and Gemini 2.0.
- **Tasks:** variable summation, GSM8K, MMLU and HumanEval.
- **Result:**
  - Performance fell by between 13.9% and 85% within the models' claimed windows, even when retrieval was perfect.
  - With whitespace padding, Llama lost 48 points on variable summation.
  - With masked padding, the open models lost at least 7.9% at 30K tokens.
  - Closed models degraded less (GPT-4o lost about 7 points on GSM8K).
- **Limitations:**
  - All experiments were run once, and per-condition sample sizes are not clearly reported.
  - Masking removes content but keeps position distance, so "length" here partly means positional distance.
  - Masking is possible only for open-weight models.

> **Conclusion 3.1 (Moderate).** Input length degrades accuracy even when the relevant evidence is fixed and can be shown to have been retrieved. The effect is well replicated in open models of 70B or less at 30K tokens or less. It is smaller in closed frontier models, and it has not been measured with this level of control at the scale of modern windows (hundreds of thousands of tokens).

3.2 Position of the relevant information

Liu et al., "Lost in the Middle" (TACL 2024)

- **Setup:** The position of a gold passage was moved among 10–30 retrieved distractors, about 1.5K–4.4K tokens. There was also a synthetic key–value retrieval task.
- **Design strength:** Two reference points frame the results:
  - *Closed-book:* no documents at all.
  - *Oracle:* the gold document only.
- **Result:** For GPT-3.5-Turbo with 20 documents:
  - 75.8% with the gold passage first;
  - 53.8% at the worst middle position;
  - 56.1% closed-book;
  - 88.3% oracle.

Burying the evidence in the middle therefore did *worse than giving no context at all*. The curve is U-shaped, with primacy and recency advantages.

Later work makes this effect conditional rather than universal.

- **Veseli et al. (2025, preprint)** varied input length as a fraction of each model's window, from 0.06 to 1.0, across six open models.
  - The U-shape was strongest when inputs filled up to about 50% of the window.
  - Beyond that, the primacy advantage disappeared and only recency remained.
  - Once the analysis was restricted to items where retrieval succeeded, the positional bias in reasoning largely flattened.
  - The authors suggest this reconciles a strong mid-context effect in short-input studies with its weakness in benchmarks above 100K tokens.
- **Chroma's Context Rot report (2025, vendor technical report):** across 18 models, needle position had no notable effect, while length and distractors did.

- **LaRA (ICML 2025):** long-context accuracy still dropped for mid-context answers. Retrieval-augmented reading did not show this drop.
- **Interventions also shrink the U-shape:**
  - training on data that requires using information anywhere in the input (An S. et al., NeurIPS 2024);
  - calibrating positional attention bias at inference time (Hsieh C.-Y. et al., Findings of ACL 2024).

Both suggest the effect comes from how models are trained and how attention is biased, not from a fixed limit of the architecture.

> **Conclusion 3.2 (Strong for existence; Moderate for scope).** Position effects are robust at short lengths and partial window fill. At high window fill, primacy weakens and recency dominates. The bias is reduced by targeted training and calibration. A mid-context failure in one model, at one length, does not predict the same failure elsewhere.

3.3 Distractors, similarity and literal matching

Intuition. A long context is rarely just noise. It contains material that looks relevant but is not. Several designs show that this resemblance, more than raw length, decides how badly models fail.

Short-context baseline: Shi et al. (ICML 2023, GSM-IC)

- Adding a single irrelevant sentence to easy GSM8K problems reduced the share of base problems solved under *every* distractor variant to below 30%.
- The best "macro accuracy" (solved under all variants) was about 45%.
- Distractors on the same topic, or containing numbers in the problem's range, were the most damaging.

Retrieval-augmented settings: Jin et al. (ICLR 2025)

- As more passages were retrieved, output quality first rose and then fell, even though retrieval recall kept increasing.
- The main cause was "hard negatives": passages that score as relevant but do not contain the answer.
- Higher retriever recall therefore does not guarantee higher accuracy.

Literal matching: NoLiMa (ICML 2025)

- **Setup:** Needle-in-a-haystack (NIAH) tests where the needle and question share minimal vocabulary. Linking them requires one or two steps of association. 13 models with windows of 128K or more.
- **Result:**
  - At 32K, 11 of the 13 models fell below 50% of their own short-context score.
  - GPT-4o went from 99.3% to 69.7%.
  - Supplying literal cues as multiple-choice options raised Llama-3.3-70B's 2-hop accuracy at 32K from 25.9% to 87.2%. This shows that word overlap is what makes standard NIAH easy.
  - Reasoning models also fell below half their base score on the hard subset.

Similarity and coherence: Context Rot (Chroma, 2025)

- Lower similarity between needle and question made performance fall faster with length.
- One distractor hurt, and four hurt more.
- Shuffled haystacks scored *better* than coherent ones in all 18 models.
- Model families differed in failure style: some abstained more, others hallucinated more.
- Chroma sells retrieval infrastructure, which should be kept in mind.

2026 frontier models: PredicateLongBench (Jain & Velingker 2026, preprint)

- At a *fixed* 128K tokens, enlarging the search space dropped one frontier model's accuracy from 92% to 10%.

- Decoys scattered through the context caused collapse. The same decoys clustered near the target did not.
- This points to dispersion and decoys as the main difficulty, rather than token count.

Conclusion 3.3 (Strong). Distractors that resemble the target, and low lexical overlap between question and evidence, are major causes of failure inside long contexts. Their effect grows with length. Benchmarks that let a model find evidence by literal matching overstate reliability.

3.4 What saturated benchmarks miss

Vanilla NIAH no longer discriminates between models.

- Vendor reports cite near-perfect single-needle recall at very large windows. For example, the Gemini 1.5 report gives above 99.7% up to 1M tokens.
- The same report shows much lower scores on harder variants:
  - about 70% for 100 needles at 128K;
  - a little above 60% at 1M.
- The report itself calls single-needle NIAH a simple, prompt-sensitive test.

Prompt sensitivity is large.

- Anthropic reported that one model's single-needle accuracy rose from 27% to 98% after adding a short prefill to the response.
- Machlab and Battle (2024, preprint) found that changing only the needle and question moved one model between 100% and 68.2%.
- Needles that conflict with what the model already knows did worst, because the model fell back on its prior knowledge.

Synthetic scores do not track realistic tasks well. HELMET (ICLR 2025)

- Tested 59 models across seven application categories.
- Across 35 instruction-tuned models at 128K, the Spearman correlation between synthetic recall scores and downstream tasks never averaged above 0.8.
- Retrieval-augmented QA tasks were the best single proxy.

Length is often confounded with difficulty in realistic benchmarks. LongBench v2 (ACL 2025)

- Scores are not monotone in length. For one leading model, "long" items scored higher than "medium" ones.
- The length buckets contain different documents and different questions, so length is confounded with how hard the question is. Human expert scores were also non-monotone (47.2 / 59.1 / 53.7).

Conclusion 3.4 (Strong). Near-perfect single-needle recall is not evidence of reliable fact use at that length. Tasks that need aggregation, tracking, multi-hop linking or resistance to decoys show effective lengths well below advertised windows (RULER; SYNC; Michelangelo; BABILong). Realistic benchmarks need length-matched designs before a length effect can be read from them.

3.5 Proposed internal mechanisms

Several accounts converge. They are mostly tested on open models of 7–70B parameters.

Architectural position bias. Wu et al. (ICML 2025) analyse attention masks theoretically.

- Causal masking pushes deeper layers toward early tokens.
- Relative positional encodings such as RoPE add distance decay.
- The interaction between the two can produce primacy and attention sinks.

Under-trained long distances. An et al. (ICLR 2025) argue that long relative positions are rare during training.

- A training-free shift of position indices (STRING) improved RULER scores by about 15 points for one 70B model.
- This is an interventional result. It supports the account but does not exclude alternatives.

"Knows but doesn't tell." Lu et al. (Findings of EMNLP 2024) trained layer-wise linear probes.

- The probes located mid-context targets more accurately than the models' generated answers did.
- Hsieh C.-Y. et al. found that calibrated attention ranks the gold document far better than raw attention.

Together these suggest that, in these models, information is often *encoded* but not *used*.

Copying circuits. Wu W. et al. (ICLR 2025) identified sparse "retrieval heads", under 5% of heads.

- Pruning them caused retrieval failures and hallucination. Pruning random heads did not.
- This gives a causal, component-level signature for "failed to copy from context" in open models.

> **Conclusion 3.5 (Moderate for open models; inferred only for closed models).** Position bias, under-training at long distances, and failure to use encoded information are all supported by interventional evidence in open models. Whether the same mechanisms explain failures in closed frontier models has not been shown.

3.6 Boundary conditions and counterevidence

Long context does not always lose to retrieval. With enough compute, long-context reading beat retrieval-augmented generation (RAG) on average for three proprietary models (Li Z. et al., EMNLP 2024 Industry). LaRA (ICML 2025) found the winner depends on conditions: long context won at 32K, for strong models and on comparison tasks; RAG won at 128K, for weaker models and at detecting that the answer is absent. Closed models degraded far less than open ones in Du et al., and in SYNC (EMNLP 2025) one closed model held 93–96% on a graph task where an open 70B model fell from 98.7% to 11.3% between 32K and 128K. Degradation is not monotone everywhere: Michelangelo (Google DeepMind preprint) found a sharp early drop, often by 32K, after which some models flattened up to 1M tokens. Most controlled evidence covers lengths of 32K or less, open models from 2023–2025, and single runs without confidence intervals; frontier-model evidence from 2026 is sparse and mostly preprint.

4. Q1b: Multi-turn accumulation within a session

Intuition. A long conversation differs from a long document in two ways:

- The information arrives in pieces, and the model often answers before it has all of them.
- Much of the context was written by the model itself, so earlier mistakes become part of what it later reads.

Either difference, rather than length as such, could explain degradation.

4.1 Underspecified, piecewise disclosure

Laban et al. (ICLR 2026): "LLMs Get Lost in Multi-Turn Conversation"

- **Setup:**
  - About 600 instructions across six task types: code, SQL, API actions, math, data-to-text and summarisation.
  - Each instruction is split into "shards." A simulated user reveals one shard per turn.
  - More than 200,000 conversations across 15 models from eight families, with 10 simulations per model and condition.
- **Conditions:**

Condition	What the model sees	GPT-4o score (example)
FULL	The whole instruction in one turn	93.0
CONCAT	All shards as one single-turn list (control for content and rephrasing)	90.9
SHARDED	One shard per turn, multi-turn and underspecified	59.1
RECAP / SNOWBALL	Sharded, with shards restated at the end or every turn	76.6 / 65.3

- **Result:**
  - Across models, SHARDED performance averaged about 35–39% below FULL.
  - CONCAT kept about 95% of FULL, which rules out loss of information content or rephrasing as the cause.
  - The authors split the drop into two parts:
    - *aptitude*, the 90th-percentile score, fell about 16%;
    - *unreliability*, the spread between 90th- and 10th-percentile runs, rose about 112%.
  - Setting temperature to 0 cut unreliability by 50–80% for single-turn conditions but not for SHARDED, where about 30% unreliability remained.
  - In a sub-study varying the number of shards from 2 to 8, degradation appeared from 2 shards onward but did not grow steadily with more.
  - Observed behaviours:
    - answering early, on assumptions;
    - building on earlier wrong attempts;
    - over-weighting the first and last turns;
    - verbosity.
- **Limitations:**
  - The simulated user may differ from real users. The authors argue it probably underestimates degradation.
  - All tasks are analytical, in English.
  - No length-matched, fully specified multi-turn control was run. As a result, underspecification is confounded with turn count.

Underspecification as the main driver. Two 2026 preprints treat the gap as behavioural and recoverable:

- An intent-rewriting mediator recovered roughly 55–62% of the gap for frontier models (Liu et al. 2026).
- Training a model to defer and ask for clarification, plus self-distillation, recovered 92–100% of single-turn accuracy on the math task (Chen, Wu and Leskovec 2026).

Both are unreviewed and use small or cooperative test settings.

> **Conclusion 4.1 (Strong for the effect; Moderate for the cause).** Revealing a request across several turns causes large drops and much greater run-to-run variability. Delivering the same content in one turn does not. The best-supported cause is underspecification combined with premature commitment, not turn count or token count.

4.2 Conditioning on self-generated errors

Three designs isolate what the model's own earlier turns contribute.

Gold-history swap: MT-Eval (Kwan et al., EMNLP 2024)

- In recollection tasks, replacing the model's earlier answers with correct ones raised judge scores substantially. For example, Mixtral went from 3.21 to 7.11 on a 10-point scale.
- In an audit of 160 low-scoring responses, about 48% had been misled by errors that accumulated earlier in the conversation.

Injected error rates: Sinha et al. (2025/2026, preprint)

- A synthetic task: look up values and keep a running sum, with the plan and the knowledge supplied in the prompt.
- Artificial assistant histories were injected with controlled error rates.
- With an error-free history, accuracy at turn 100 was still below turn 1. That residual is the pure long-context component.
- Raising the injected error rate lowered accuracy further. That is the self-conditioning component.
- Scaling to larger models did not remove self-conditioning. The "thinking" models tested were largely immune.

Length-matched neutralisation: Li et al. (2026, preprint)

- Replayed the SHARDED conversations, replacing earlier assistant turns with "Acknowledged."
- A length-matched version padded the placeholder to the original turn's length.
- Short and length-matched neutralisation produced the same effect, with a difference of −0.001 (95% CI −0.039 to 0.035). Shortening the context is therefore not what drives it.
- The effect was mixed:
  - neutralisation rescued about 51% of failures but broke about 15% of successes;
  - the net macro effect was small (+0.027);
  - some models and tasks got worse.

Extra turns without new information. Huang et al. (ICLR 2024) show that turns adding no new information can hurt. Unprompted self-correction lowered GPT-3.5's CommonSenseQA accuracy from 75.8% to about 40% over two rounds.

> **Conclusion 4.2 (Moderate).** The content of the model's own earlier turns is a separate, model-dependent cause of multi-turn degradation, distinct from length. It mostly harms but sometimes helps. Three independent designs agree on its existence. They disagree on its average size.

4.3 Interference when facts are updated within a session

PI-LLM (Wang & Sun 2025, preprint)

- **Setup:**
  - A stream of key–value updates is presented, and the model must report the latest value for each key.
  - The answer is always the most recent update, so finding it is trivial.
  - Updates per key were varied from 3 to 400.
- **Result:**
  - Accuracy fell roughly log-linearly toward zero as the number of updates grew.
  - Most errors were *superseded* values: the model reported an older value instead of the latest one.
  - In a length-controlled experiment, total input length was held fixed while the number of tracked keys varied. The decline remained.
  - Across 30 models, parameter count predicted resistance to this interference ($p = 0.005$). Context-window size did not ($p = 0.886$).
  - Chain-of-thought did not help.
  - Natural-language instructions to "forget" helped by less than 10 points.

This bears directly on conversations in which a user corrects a fact. Of all within-session findings, it maps most closely onto the *update* failures seen in cross-session memory (Section 5).

> **Conclusion 4.3 (Moderate; single preprint, but with a length control).** Interference from superseded values degrades in-context fact tracking independently of total length. It may be a shared factor behind update failures in both mechanisms.

4.4 Social pressure in user turns

In FlipFlop (Laban et al. 2023/2024, preprint; 67,640 two-turn experiments, 10 models, 7 tasks), a single user challenge made models change their answer about 46% of the time and cut accuracy by about 17%. SYCON-Bench (Hong et al., preprint) found that under five turns of repeated pushback, larger models gave way less often and alignment tuning was associated with giving way more; only one of its three scenarios concerns factual premises. This is a separate cause that a single turn can trigger, independent of length, but it is easily mistaken for "the conversation got long."

4.5 Compaction and summarisation within a session: a boundary case

Assistants that summarise earlier turns as a chat nears the window limit use a lossy operation that also appears in memory systems. The evidence on this is thin.

Recursive summarisation (Wang Q. et al., Neurocomputing)

- A manual audit of 100 summarised memories found:
  - fabricated facts in 2.7%;
  - wrong relations in 3.2%;
  - missing details in 3.9%.
- The setting is cross-session (MSC), but the same error types arise in compaction within a session.

CONPACT (Wang Z. et al. 2026, preprint; abstract only)

- Compactors retained about 17% of injected side constraints on average.
- In one agent test, compaction preceded unapproved deletions in 30 of 100 sessions, against 0 before compaction.
- These are constraints, not facts.

Under the approved boundaries, such systems are assigned according to how they describe themselves. A compacting assistant that keeps the full history available for reference is within-session. A failure caused by its summary has the *character* of a memory-write failure, so the distinction is a matter of definition rather than of evidence.

4.6 Counterevidence: length alone is not always harmful

BoolQ with filler turns (Neergaard et al. 2026, preprint)

- **Setup:**
  - The target question was placed after 1–21 filler turns.
  - Three small open models were tested.
- **Result:**
  - Neutral filler turns were mostly harmless or slightly beneficial.
  - Harm appeared only when misleading content was combined with length.

Drift over turns (Dongre et al. 2025, workshop)

- Behavioural drift settled at bounded equilibria rather than growing without limit.

Turn counts in the sharding study (Laban et al.)

- Degradation did not keep growing as the number of shards increased.

Overall. No verified study fully crosses turn count, token length, level of specification, and correct versus self-generated history. Elements of that design exist across Laban et al., Sinha et al., Li et al. (2026) and Neergaard et al., but no single study combines them.

> **Conclusion 4.6 (Moderate).** "Conversation length" is not a single cause. The large multi-turn effects come from how information is disclosed and from the model's own history. Pure length adds a smaller residual. Neutral turns can be harmless, and misleading turns are not.

5. Q2: Failures of cross-session memory

Intuition. A memory system can go wrong at four points:

- **Write:** it may never store a fact, or store it wrongly. This includes stamping it with the date it was *mentioned* rather than the date it *happened*.
- **Retrieve:** it may fail to bring the fact back when it is needed.
- **Read:** it may bring the fact back and the model may still misuse it.
- **Update:** it may keep using a fact after that fact has stopped being true.

Each step can fail on its own, and the error a user sees looks the same whichever step caused it.

5.1 The main benchmarks

Benchmark	Venue / status	History size	What it probes	Metric / judge
MSC (Xu, Szlam & Weston 2022)	ACL 2022	Up to 5 short sessions	Open chit-chat continuity	Perplexity, human rating; not factual QA
LoCoMo (Maharana et al. 2024)	ACL 2024	10 conversations, ~27 sessions, ~16.6K tokens	Single- and multi-hop recall, temporal, false-premise QA	Token F1; human ceiling 87.9
LongMemEval (Wu et al. 2025)	ICLR 2025	S: ~115K tokens; M: ~1.5M tokens; oracle	Extraction, multi-session, temporal, knowledge update, abstention	GPT-4o judge, ~97% agreement with experts
MemoryAgentBench (Hu et al. 2026)	ICLR 2026	Up to ~1.4M tokens, fed in incrementally	Retrieval, learning at test time, long-range understanding, selective forgetting	Accuracy / F1 / substring match
BEAM (Tavakoli et al.)	Preprint; ICLR 2026 listing not opened	100K–10M tokens	10 abilities incl. contradiction resolution and event ordering	Nugget-based LLM judge; no human validation found
PersonaMem (Jiang et al. 2025)	NeurIPS 2025 workshop	Up to 60 sessions	Tracking preferences as they change	Multiple choice
DialSim (Kim et al. 2024)	Preprint	~350K tokens per show	Real-time QA over TV scripts; checks for prior knowledge of the show	Accuracy; oracle condition
TReMu (Ge et al. 2025)	Findings of ACL 2025	Built on LoCoMo	Temporal anchoring, precedence, intervals	Multiple choice

5.2 Best-supported failure modes

Updating and contradiction resolution (Strong across groups, but mostly synthetic data)

Study	Finding
MemoryAgentBench (Hu et al.)	Every method, whether long-context, retrieval or agentic memory, reached at most about 7% on multi-hop conflict resolution.
Same study, solvability check	One reasoning model scored 80.0 at 6K tokens and 14.0 at 32K. The task can be solved, but performance collapses as length grows.
BEAM	Contradiction resolution was the lowest-scoring ability for every method.
PersonaMem	Frontier models were around 50% when they had to track preferences that changed over time.
LongMemEval error analysis	On knowledge-update questions, retrievers often found the new fact but not the old fact it replaced.

Temporal reasoning across sessions (Strong)

- LoCoMo: the best long-context model scored temporal F1 51.4, against 92.6 for humans.
- TReMu: GPT-4o reached about 30% with standard prompting and about 62% with chain-of-thought.
- Design changes help, which locates where the failure lies:

- Tagging each memory with the inferred date the event happened, rather than the date it was mentioned, and computing dates with code, raised TReMu to about 78%.
- Time-aware query expansion raised temporal retrieval recall by about 7–11% (LongMemEval).
- Bi-temporal records that invalidate old facts instead of deleting them raised temporal accuracy from 45.1 to 62.4 in a vendor-authored evaluation (Zep).

Multi-session aggregation and multi-hop questions (Moderate)

- LongMemEval multi-session questions remain weak even with oracle sessions.
- DialSim: two-hop questions over a temporal knowledge graph scored 13.5%, against 66.7% for one-hop questions.

False premises and abstention (Moderate)

- On LoCoMo's adversarial questions, F1 *fell* as context grew. One model dropped from 34.8 at 4K to 14.8 at 16K, which the authors read as more hallucination with longer inputs.
- LongMemEval tests abstention separately with 30 false-premise items. That small number makes the per-type estimate noisy.

Write-stage loss (Moderate)

- In LongMemEval, storing only extracted facts instead of raw rounds *lowered* overall QA accuracy. Extraction discards information.
- In Zep's evaluation, facts that the assistant itself had stated were stored less well than user facts. Single-session-assistant accuracy fell from 94.6 to 80.4.
- In a longitudinal synthetic instrument (Spencer 2026, preprint), weakly written facts failed 24.2% of the time, against 1.6% for clean writes. Recall of early facts declined as a size budget evicted them.

Applying memory that no longer applies (Limited; one preprint with a judge cross-check)

- MemTrapBench (Wang et al. 2026) plants a prior fact, buries it, and then asks a question under changed conditions where that fact no longer holds.
- Every memory strategy scored *below* no memory. For one model: 85.2% with no memory, against a best of 71.2% with memory.
- Control histories without traps did not hurt.

Deployed assistants

- LongMemEval ran a small probe in 2024: 97 questions over short histories of 3–6 sessions, entered through the live products.
- Two commercial assistants with memory features scored 37% and 64% below the same underlying model reading the same history offline.
- This shows *that* failures occur in deployed memory pipelines. It cannot show *where* they occur, because the pipelines are opaque.

5.3 Which stage fails?

LongMemEval treats memory as three stages: *indexing* (what is stored and under which key), *retrieval* (how the query is formed) and *reading*.

Oracle versus full history (accuracy):

Model	Oracle sessions	Full history (S)	Drop
GPT-4o	0.870	0.606	−30%
Llama-3.1-70B	0.744	0.334	−55%

Retrieval versus reading. Errors split roughly evenly:

- About 15–19% of all instances were cases where retrieval was correct and the answer was still wrong.
- Correct answers almost always depended on correct retrieval.

Reading strategy alone: changing how the model reads retrieved material (Chain-of-Note, structured JSON input) moved accuracy by up to about 10 points, even with perfect retrieval.

Yuan, Su & Yao (2026, preprint) ran a 3×3 factorial on LoCoMo with one model (GPT-5-mini).

- **Factors:**
  - *Write format:* raw chunks; extracted facts, in the style of one commercial framework; session summaries, in the style of another.
  - *Retrieval method:* cosine similarity; BM25; a hybrid with LLM re-ranking.
- **Results:**
  - The retrieval method accounted for about 20 points of variance. The write format accounted for only 3–8 points.
  - Raw chunks matched or beat LLM-extracted formats.
  - Retrieval precision tracked accuracy at r = 0.98.
  - Failure types:
    - retrieval failure: 11–46% of questions;
    - failure to use memory that had been retrieved: about 4–8%;
    - hallucination: about 0.4–1.4%.
  - Judge validation against humans: κ = 0.82.
- **Limitations:** one model, one benchmark, prompt-based reimplementations of the write formats, and no full-context arm.

How the two decompositions differ.

- LongMemEval finds a near-even split between retrieval and reading. Yuan et al. find retrieval dominant.
- These are not straightforward contradictions. The two studies differ in:
  - benchmark: about 115K tokens versus about 16K;
  - model;
  - how a "use failure" is coded;
  - the retrieval configurations being compared.
- Both agree that *write* format matters less than retrieval quality on short histories. The write-stage findings in §5.2 show it still matters for particular fact types and over long periods of use.

5.4 Do memory systems beat simply reading the whole history?

This question decides whether a benchmark exercises mechanism (b) at all. The evidence is contested.

LoCoMo histories fit inside modern windows, and full-context baselines win there.

- A vendor's own paper (Chhikara et al. 2025, Mem0) found full context had the highest judged accuracy (J = 72.90). All memory systems tested scored lower. The headline gains it claimed were in latency and token cost.
- An independent audit of the LoCoMo answer key (2026) reported:
  - 6.4% of gold answers are wrong, capping attainable accuracy near 93.6%;
  - the LLM judge accepted 62.81% of answers that were deliberately wrong but vague;
  - a full-context baseline with a chain-of-thought prompt scored about 92.6%.
- Rival vendors disputed one another's harnesses. Disputed choices included how speaker roles were assigned and where timestamps were placed, and one vendor could not reproduce another's setup. Their figures for the same system differ by about 9 points.

On LongMemEval-S (~115K tokens) the evidence is mixed.

- A vendor-authored evaluation (Zep) reported 71.2% against 60.2% for full context with GPT-4o, while cutting context from about 115K to 1.6K tokens. It also lost accuracy on facts the assistant itself had stated.
- Single-factor re-analyses found parity:

- MemDelta (2026 preprint): retrieval at 47.2% versus full context at 49.8%, not significant ($p = 0.34$).
- Swapping only the embedding model changed accuracy by 6.2 points, more than the gap between architectures.
- Which answer model was used could flip the winner. In one model, about 63% of full-context failures were refusals.

Why this matters. Jiang et al. (2026, preprint) propose a "context saturation gap": memory-system accuracy minus full-context accuracy. They note that LoCoMo and MemBench fit within current windows, and only LongMemEval-M clearly requires external memory. On a benchmark where everything fits in the window, a "memory failure" is largely a long-context reading failure.

> **Conclusion 5 (Moderate overall).** Cross-session memory has its own failure modes that are well supported: update, temporal reasoning, aggregation, abstention, write-stage loss and over-application. Their *relative* size against long-context failures depends on whether the history fits in the window, which retriever and reader are used, and how judging is done. Leaderboard-style comparisons of memory systems are not reliable evidence. Many are vendor-run, single-run, judged by lenient LLM judges, and disputed.

6. Q3: Evaluation designs that can separate the mechanisms

Intuition. To blame a failure on a cause, an experiment must change one factor and hold the rest constant. The table below lists the designs found, what each one holds constant, and what it leaves uncontrolled.

Design family	Examples	Controls for	Leaves open
Length-only manipulation at a fixed task and fixed evidence	Levy et al.; Du et al. (whitespace, masking, recitation check); NoLiMa; 6K-vs-32K check in MemoryAgentBench	Content of the evidence; retrieval success (Du); lexical shortcuts (NoLiMa)	Which internal cause is responsible; padding that is out of distribution; generalisation to natural text and to windows of hundreds of thousands of tokens
Oracle / gold evidence versus full history	LongMemEval oracle vs S/M; Liu et al. closed-book and oracle anchors; Context Rot focused (~300 tokens) vs full (~113K); DialSim oracle	Answerability; estimates the cost of processing for a fixed set of facts	The oracle removes both length *and* the retrieval burden. It does not test the write stage: evidence is supplied verbatim, so storage is never exercised
Swapping the route: retrieval vs long context, per query	Xu et al. (ICLR 2024); Self-Route agreement analysis; LaRA; order-preserving RAG	Model and query; attributes a failure to retrieval or chunking where the long-context route succeeds	The routes differ in length, ordering and noise all at once; when both fail, nothing can be attributed
Stage-wise decomposition	LongMemEval indexing/retrieval/reading; recall@k vs accuracy; RAGChecker claim-level metrics; Barnett et al. failure points; Yuan et al. factorial	Which pipeline stage failed	A fact retrieved but not used cannot be split into neglect caused by length versus a reasoning error without an added length control; LLM-based checkers have imperfect agreement
Structure-matched controls	Laban et al. CONCAT vs SHARDED	Information content versus how it is disclosed across turns	Simulated users; turn count confounded with underspecification
Manipulating the model's own history	MT-Eval gold-history swap; Sinha et al. injected error rates; Li et al. length-matched neutralisation	Self-conditioning versus length	Synthetic or replayed settings; the user does not adapt

Design family	Examples	Controls for	Leaves open
Update-targeted and time-targeted conditions	MemoryAgentBench selective forgetting; PI-LLM; TReMu; MemTrapBench; Spencer's write-quality and tenure probes	The update and write stages, i.e. mechanism (b) proper	Mostly synthetic data; small samples; rankings that depend on the judge
Intervention-based internal attribution	Retrieval-head pruning; ContextCite sentence ablation; layer-wise probes	Causal claims about which components or which input spans produced an output	Requires access to weights or log-probabilities; linear or probe-specific assumptions; not available for closed deployed systems

Three points stand out.

1. The oracle condition is the single most informative design for separating the two mechanisms. Its gap has a specific meaning, though. In LongMemEval, the oracle-versus-S gap measures the cost of *finding and reading* the evidence inside a long history. That is mechanism (a) together with retrieval in context. It does not measure whether a memory system would have *stored* the evidence. Only update-targeted, write-targeted and longitudinal designs reach the store itself, and they are the least mature group.

2. Length controls and oracle conditions are rarely combined. Du et al. combine a recitation check, which confirms retrieval, with padding that adds no content. MemoryAgentBench combines conflict resolution with a short-versus-long check. These are the closest the literature comes to a full separation. Both still cover few models.

3. Validity critiques limit what any one design can establish.

- Literal matching inflates scores on needle tests (NoLiMa).
- Synthetic scores correlate only moderately with realistic tasks (HELMET).
- LLM judges can differ from human scores by 5–10 points while showing over 90% agreement, and they accept vague wrong answers (Thakur et al. 2024/2025; the LoCoMo audit).
- Answer keys contain errors (LoCoMo).

Effects smaller than about 5–10 judged points should be treated as unresolved unless the judge has been validated on the benchmark itself, as in LongMemEval at about 97% agreement and Yuan et al. at $\kappa = 0.82$.

7. Q4: How far can a specific failure be attributed?

7.1 What can be attributed, and at what level

Attribution is defensible at the **population level**. That requires three things:

- a named controlled contrast, such as oracle versus full history, padded versus unpadded, or CONCAT versus SHARDED;
- a stated benchmark and set of models;
- an effect larger than the measurement noise.

Under those conditions the literature supports statements such as:

- Holding evidence fixed, this model loses N points from added length.
- About half of this system's errors on this benchmark occur after correct retrieval.
- Revealing the request across turns, rather than the information content, accounts for this drop.

For **a single observed failure** in a deployed assistant, attribution is rarely possible. The reasons are set out below.

7.2 Limits on attribution

1. The mechanisms are nested, not parallel.

- Retrieved memories are placed into a prompt. A cross-session error can therefore come from retrieval failing, or from the reader losing a memory that was correctly retrieved. The second case is mechanism (a) operating inside (b).
- LongMemEval's gains from reading strategy under perfect retrieval show this interaction directly. So does RAGChecker's finding that higher retriever recall *increases* a generator's sensitivity to noise.
- Because of this nesting, results from single-factor designs cannot simply be added up into a complete breakdown.

2. Effects are non-monotone and interact.

- Several studies find inverted-U curves as more is retrieved: Jin et al.; order-preserving RAG; the Databricks long-context RAG study.
- LaRA finds interactions among model, length and task.
- A cause identified in one configuration therefore does not carry over to another.

3. Outputs vary from run to run.

- Laban et al. found unreliability of about 30% in multi-turn settings even at temperature 0.
- A single failure may simply be sampling variation. Attributing it requires repeated trials, and most benchmark papers report single runs.

4. Measurement noise is comparable to the effects.

- Judge–human gaps of 5–10 points; acceptance of vague wrong answers; about 6% of gold answers wrong; rankings that change with the judge (Spencer 2026).
- Each of these can be as large as the difference being attributed.

5. Deployed systems are opaque.

- From the outside, nobody can see:
  - whether a memory was written;
  - what was retrieved;
  - how the context was assembled;
  - whether compaction occurred;
  - which model version, or which serving configuration, handled a turn.
- Vendor documentation states that provenance displays are incomplete (Section 8).

6. Model self-reports are not valid evidence.

- Turpin et al. (NeurIPS 2023) showed that chain-of-thought explanations can leave out the factor that actually drove the answer.
- An assistant saying "I couldn't find that in our earlier chats", or citing a memory, does not establish which mechanism failed.

7. Benchmarks sometimes measure a different mechanism from the one they name.

- When a "memory" benchmark fits within the context window, failures on it are largely failures of long-context reading (Section 5.4).
- When a needle test can be solved by literal matching, success on it is not evidence of how the model uses context (NoLiMa).

7.3 An attribution ladder

This is a description of evidential strength, not a recommendation about mitigations. The approved scope excludes such recommendations.

Level	What it takes	What it supports
0. Observation	A single failure in a deployed assistant	Nothing about mechanism
1. Reproduction	Repeated trials with the same input	That the failure is systematic rather than sampling noise
2. Route contrast	The same question answered with the full history in context and with retrieval	Whether finding the evidence or reading it is implicated, within the confounds noted for route substitution

Level	What it takes	What it supports
3. Oracle and length contrast	Gold evidence alone, then gold evidence with length-matched padding	Separation of retrieval, raw length and reading failures
4. Stage instrumentation	Logs of what was written, what was retrieved and what was assembled into context	Attribution to the write, retrieve or read stage
5. Internal intervention	Ablating components or spans with access to weights or log-probabilities	A causal mechanism in that model

Levels 2–3 can be run by outside researchers on models available through an API. Levels 4–5 need the system's internals. In practice, only the vendor or an open-weight replication can reach those levels for a deployed assistant.

8. Q5: Controlled evidence compared with reports about commercial products

How to read this section. Everything here belongs to the anecdotal evidence class. It is included to show how product reports relate to the controlled findings. It is not evidence that any mechanism caused any reported failure, and it does not compare or rank products. Vendors are named only so the sources can be traced.

8.1 What vendor documentation concedes

Public documentation of consumer memory features from OpenAI, Anthropic, Google and Microsoft, collected in 2026, describes several limits without any external prompting:

Selective storage and retrieval

- Memory does not keep every detail of past chats.
- Chat-history reference is not a fixed list of facts.
- When memory is full, entries can be deprioritised by recency or frequency.

Deletion lag

- Deleted memories, or memories from deleted chats, can keep being referenced for days. One vendor documents up to seven days.
- Turning memory off may stop it being applied without deleting what is stored.

Incomplete provenance

- Source displays may show only some of what influenced a reply.
- A "saved info" label can sometimes be wrong.
- One enterprise product documents that memory writes produce no audit-log entries.

Compaction within a session

- At least one product automatically summarises earlier messages as a chat nears its window limit, while keeping the full history available for reference.
- What the summary keeps is not documented.

Effective context below the nominal window

- One vendor's engineering guidance names "context rot": recall and reasoning over details weaken as the window fills.
- It describes context as a finite attention budget.
- *Read through secondary summaries; the primary page was not reachable.*

Version churn

- Memory designs changed several times between 2024 and 2026 at each vendor. Examples:
  - adding reference to past chats;
  - moving from a periodically refreshed summary to topic-level memories;
  - a background "synthesis" process.
- A product report is therefore a snapshot of one configuration.

What is not documented is what attribution would need:

- how retrieval ranks memories;
- how conflicting memories are resolved;
- how many tokens injected memory receives;
- when compaction triggers;
- whether stored items are verbatim or summarised.

8.2 Vendor-reported evaluations

Several vendors publish long-context figures that repeat the controlled pattern: near-perfect recall of a single needle alongside much weaker results on harder variants.

- **Gemini 1.5 technical report (2024)**
  - Single-needle recall above 99.7% up to 1M tokens.
  - About 60–70% on 100-needle recall.
  - About 80% on multi-round co-reference at 1M.
- **GPT-4.1 announcement (2025), as reported by secondary sources**
  - A multi-round co-reference score falling from about 80% at short context to about 50% at the full 1M window.
  - A graph-walk task falling from 61.7% to about 19% beyond 128K.
  - *These figures should be checked against the primary page before reuse.*
- **Meta's Llama 4 release (2025)**
  - Advertised a 10M-token window for a model trained at 256K.
  - Independent long-fiction scores at about 120K tokens were low.
  - The vendor attributed early results to deployment problems. That attribution is disputed and unverified.
- **OpenAI's memory "synthesis" ("dreaming") announcement (June 2026), via secondary reporting**
  - Internal task-success rates rose across three memory generations:
    - factual recall: 41.5% → 67.9% → 82.8%;
    - handling time-sensitive context: 9.4% → 52.2% → 75.1%.
  - Read as the vendor's own data, this concedes that earlier memory versions usually failed to update time-sensitive facts. That matches the temporal and update failures in Section 5.
  - The evaluation design is undisclosed and has not been independently replicated.

These are vendor-built benchmarks. They are consistent with the controlled literature but are not independent of it, and the authors of the systems ran them.

8.3 User reports and journalism

Recurring kinds of report, 2024–2026:

Reported pattern	Example source (type)	Resembles controlled failure mode	Why attribution fails
Memories silently lost or overwritten when capacity filled	Developer-forum threads, Feb 2025; journalism Nov 2025 (TechRadar)	Write/storage loss (§5.2)	No vendor statement; unknown scale; journalists judged the issue not widespread; secondary articles repeat unsourced statistics
Content from one chat surfacing in an unrelated chat	Developer forum, Mar 2025	Retrieval returning the wrong item	Unclear which memory subsystem was active at the time
Assistant recites a saved rule but does not apply it; claims to remember past chats while using only the current message	Developer forum, Aug–Oct 2026, with a vendor-support reply offering unconfirmed workarounds	Read-stage failure; confabulated continuity	Vendor neither confirmed nor explained the behaviour; model routing unknown
False memories planted through untrusted content	Security research reported in Sep 2024	Write-stage corruption	Demonstrated in research, not observed in the wild; partially fixed

Reported pattern	Example source (type)	Resembles controlled failure mode	Why attribution fails
Long-chat quality loss traced by the vendor to serving infrastructure	Vendor postmortem, Sep 2025: some requests misrouted to servers configured for a larger context window, plus two overlapping bugs	Looks like long-context degradation, but was a serving configuration	Users could not know which server handled their request; overlapping bugs confound any single report
Memory interacting with model behaviour (amplified sycophancy)	Vendor postmortem, May 2025	Social-pressure effects (§4.4), not factual recall	No data released

No vendor postmortem was found that attributes a specific factual error, given to a user, to memory retrieval or to truncation of a long chat.

8.4 Where the two evidence classes meet, and where they do not

Alignment

- Product reports and vendor documents describe the same *categories* of failure as the controlled literature:
  - effective context smaller than the window;
  - stale or unupdated facts;
  - lossy summarisation;
  - retrieval of the wrong item;
  - confident continuity without retrieval.
- A controlled study, LongMemEval, found deployed memory pipelines losing 37–64% against offline reading on the same histories. That is consistent with users experiencing memory errors on short histories.

Non-transfer from controlled evaluations to products

- Controlled results come from API models with known inputs.
- Products add steps that no controlled study observes:
  - undocumented pipelines for retrieval, summarisation and context assembly;
  - routing across models;
  - safety layers;
  - frequent version changes.
- A benchmark score for the underlying model does not predict how a product behaves.

Non-transfer from products to controlled conclusions

- User reports are self-selected and of unknown scale.
- Their symptoms can come from serving bugs or model swaps that look like context or memory failure.
- They cannot test any mechanistic claim, and under the approved boundaries they are not used to do so.

9. Answer to the primary question (Q0) and verdict on the premises

Q0. How do context length, conversation length and cross-session memory retrieval each relate to factual reliability, and how far can failures be attributed to one mechanism?

Context length.

- Relationship: real, and partly independent of content. Holding evidence fixed and confirming retrieval, added length still costs accuracy.
- The cost is amplified most by:
  - distractors that resemble the target;
  - low lexical overlap between question and evidence;

  - evidence scattered through the context.
- It is modulated by position, conditionally on how full the window is.
- It is substantially reduced, but not removed, in closed frontier models.
- Mechanisms are supported by interventional evidence in open models only.

Conversation length.

- Relationship: large degradations, driven mainly by:
  - how information is disclosed across turns (underspecification and premature commitment);
  - conditioning on the model's own earlier errors;
  - interference from superseded facts;
  - social pressure.
- Raw token length adds a smaller residual decline.
- Neutral turns can be harmless.
- Turn count as such has not been cleanly isolated.

Cross-session memory.

- Relationship: failure modes that are its own:
  - write loss and mis-dating;
  - retrieval misses;
  - stale facts not updated;
  - stale memory applied after conditions change.
- Two of these are hard for every architecture tested: temporal reasoning and multi-hop reasoning across sessions.
- Where the history fits in the window, memory systems often do no better than reading the full history. On such benchmarks, "memory failure" is largely long-context failure.

Attribution.

- Controlled designs can separate the mechanisms at the population level. The main ones are oracle conditions, length-matched padding, stage decomposition, route substitution and manipulation of the model's own history.
- They cannot do so for an individual failure in a deployed assistant.
- The mechanisms are nested: retrieval feeds a long-context read. Separating them therefore always needs a combination of designs, not one contrast.

Premise (to test)	Verdict	Strength	Key qualifications
Longer context or longer conversations degrade factual reliability	Supported, with boundary conditions	Strong (context); Moderate (conversation)	Model-, task- and definition-dependent; multi-turn effects driven more by structure than length; neutral filler can be harmless
The two mechanisms are empirically separable	Supported in controlled settings; not for individual deployed failures	Moderate	Mechanisms are nested; no single design separates every stage; the write stage is the least studied
Effective context utilisation is smaller than the advertised window	Supported on every non-trivial benchmark found	Strong	Size of the gap depends on the definition (absolute vs relative threshold, task set); single-needle tests do not show it

10. Appraisal records of conclusion-driving studies

Condensed from the full appraisal notes. "FT" means full text read, at least in part. "Abs" means abstract or landing page only.

Study (status)	Design / controls	Models; lengths	Metric; uncertainty	Main confounds	Q	Access
Liu et al. 2024 (TACL)	Position sweep; closed-book and oracle anchors; query-aware variant	GPT-3.5/4, Claude-1.3, open 7–70B; ≤4.4K	String-match accuracy; n=2,655; no CIs	Short contexts; hard-negative distractors; obsolete models	Q1, Q3	FT

Study (status)	Design / controls	Models; lengths	Metric; uncertainty	Main confounds	Q	Access
Levy et al. 2024 (ACL)	Fixed evidence; three padding types; four placements; CoT	5 models; 250–3K	Accuracy; ~300 per point; no CIs	Short lengths; binary labels	Q1, Q3	FT
Du et al. 2025 (Findings EMNLP)	Recitation check; whitespace and masked padding; evidence placement	2 open + 3 closed; ≤30K	Task accuracy; single run	Masking keeps position distance; few models	Q1, Q3, Q4	FT
RULER 2024 (COLM)	13 synthetic tasks; absolute effective-length threshold	17 main (37 appendix); 4K–128K	Accuracy; n=500 per cell; no CIs	Length and filler confounded; threshold arbitrary	Q1, Q3	FT
NoLiMa 2025 (ICML)	Minimal-overlap needles; last-2K control; literal-cue ablation	13 models; ≤32K	Relative effective length; no CIs	Template needles; strict criterion	Q1, Q3	FT
HELMET 2025 (ICLR)	7 categories; correlation of synthetic with realistic tasks	59 models; 8K–128K	Model-based judge (κ 0.91/0.76); some seeds	Length confounded with number of passages	Q3	FT (partial)
Veseli et al. 2025 (preprint)	Relative input length 0.06–1.0; position; conditioned on retrieval	6 open models	Accuracy; 100 per dataset	Open models only; preprint	Q1, Q4	FT
Jin et al. 2025 (ICLR)	Increasing number of retrieved passages; recall vs accuracy	Several open/closed	Accuracy	Hard negatives defined by retriever score	Q1, Q3	Abs
Laban et al. 2025 (ICLR 2026)	FULL / CONCAT / SHARDED; aptitude vs unreliability; temperature	15 models; >200K conversations	Task-specific; 10 simulations per cell	Simulated user; no length-matched multi-turn control	Q1, Q3, Q4	FT (partial)
MT-Eval 2024 (EMNLP)	Single- vs multi-turn; gold-history swap; distance probe	10 mostly small models	GPT-4 judge (10-point)	Not length-matched; older models	Q1, Q3, Q4	FT (partial)
Sinha et al. 2025/26 (preprint)	Injected error rate; error-free history baseline	Frontier and open; ≤100 turns	Accuracy	Synthetic task	Q3, Q4	FT (partial)
Li et al. 2026 (preprint)	Length-matched neutralisation of assistant turns; per-turn surgery	5 models; 627 instructions	Effect with 95% CI	Retrospective; non-adaptive user	Q3, Q4	FT
PI-LLM 2025 (preprint)	Update count; fixed-length control	~30 models	Accuracy; regression	Synthetic key–value stream	Q1, Q4	FT (partial)
LoCoMo 2024 (ACL)	Long-context vs RAG; write format; human ceiling	gpt-3.5/4 variants; ~16.6K	F1; no CIs	Synthetic dialogues; 6.4% key errors (audit)	Q2, Q3	FT
LongMemEval 2025 (ICLR)	Oracle vs S/M; stage decomposition; deployed probe	5+ LLMs; 115K–1.5M	GPT-4o judge (~97% agreement); single runs	Synthetic histories; n=97 probe	Q2, Q3, Q4	FT
MemoryAgentBench 2026 (ICLR)	Incremental input; four competencies; short-vs-long check	LC, RAG, agentic; ≤1.4M	Accuracy / F1	Counterfactual edits, not natural updates	Q2, Q3	FT
TReMu 2025 (Findings ACL)	Event-date vs mention-date memories; code-based date arithmetic	GPT-4o and others	Multiple-choice accuracy	Authors evaluate own method	Q2, Q4	FT
Yuan et al. 2026 (preprint)	3×3 write × retrieval factorial; failure taxonomy	GPT-5-mini; LoCoMo	Judge κ=0.82	One model; no full-context arm	Q2, Q4	FT
Mem0 2025 (vendor preprint)	Memory systems vs full context on LoCoMo	gpt-4o-mini	LLM judge; 10 runs ± SD	Vendor-authored; harness disputed	Q2, Q3	FT
Zep 2025 (vendor preprint)	Bi-temporal graph vs full context on LongMemEval-S	gpt-4o / 4o-mini	Single runs	Vendor-authored; remote service	Q2	FT

Study (status)	Design / controls	Models; lengths	Metric; uncertainty	Main confounds	Q	Access
MemTrapBench 2026 (preprint)	Planted priors made obsolete; trap-free controls	2 models; 5 strategies	Two judges cross-checked	Synthetic	Q2	FT (partial)

11. Limitations of this review

Not systematic.

- This is a narrative review. Searches were broad but not exhaustive, and no formal risk-of-bias tool was applied.
- Five parallel sweeps gathered the evidence. Overlaps between them were reconciled by hand. Where the sweeps reported different venues or figures for the same study, the review used the version checked against the venue page.

Recency and status.

- Several important 2025–2026 findings are unreviewed preprints:
  - self-conditioning (Sinha et al.);
  - length-matched neutralisation (Li et al.);
  - the stage factorial (Yuan et al.);
  - MemTrapBench;
  - window-fill position effects (Veseli et al.).
- They are labelled as preprints, and no conclusion rests on one of them alone.
- Some 2026 preprints name model versions that could not be independently checked.

Access.

- Some studies were read in abstract only: Jin et al.; BABILong at the level of per-model results; MemBench; PersonaMem; CONPACT.
- Several long papers were read only in part.
- Figures available only in charts are given as approximate.
- Several vendor primary pages were unreachable and were read through secondary sources. These are marked in Section 8.

Narrow evidence base.

- Controlled evidence is concentrated on:
  - synthetic tasks;
  - simulated users and histories;
  - open models of 70B or less;
  - lengths of 32K or less, except for the memory benchmarks.
- Few studies report confidence intervals or repeated runs.
- Results for frontier closed models at windows of hundreds of thousands of tokens remain thin and mostly vendor-reported.

Not covered: training-data knowledge. Factual errors that come from what the model learned in training are excluded by scope. Several studies note that models fall back on parametric knowledge when evidence in context is hard to use, for example Machlab & Battle and the RULER failure analysis. The excluded mechanism therefore interacts with the included ones, and this review does not separate the two.

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Status labels: [PR] peer-reviewed (venue given); [PP] preprint; [TR] technical report; [VB] vendor blog or announcement; [VD] vendor documentation; [J] journalism; [F] forum. Product sources ([VB], [VD], [J], [F]) belong to the anecdotal evidence class.

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